Electric Potential

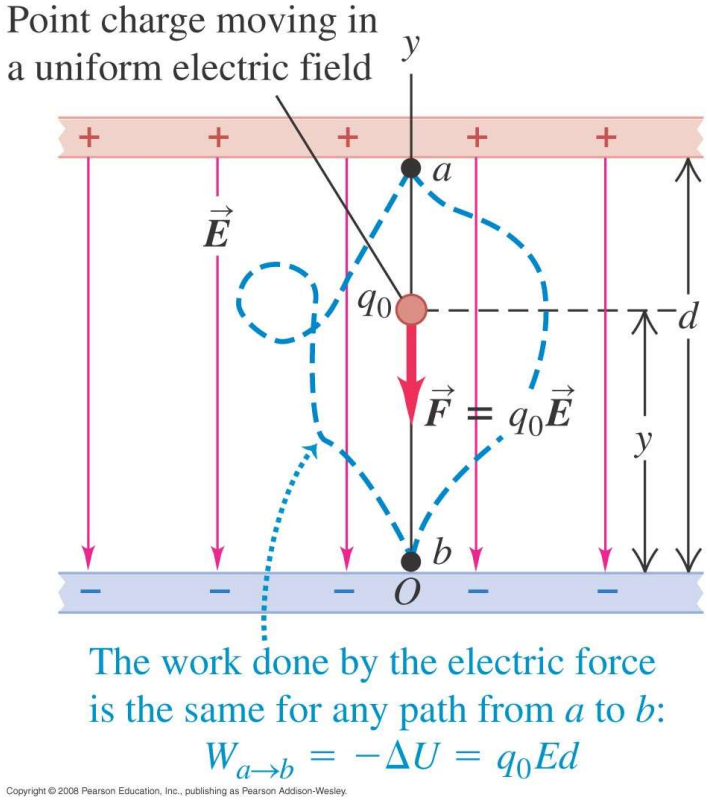
Electric potential energy – uniform field

- If the upper plate is positive, a positive charge will experience a downward force given by

$$F = q_o E$$

- The work done by the force is

$$W_{a\to b} = -q_0 E y_b + q_0 E y_a$$
$$= -U_b + U_a$$
$$= -(U_b - U_a)$$
$$= -\Delta U = q_0 E d$$

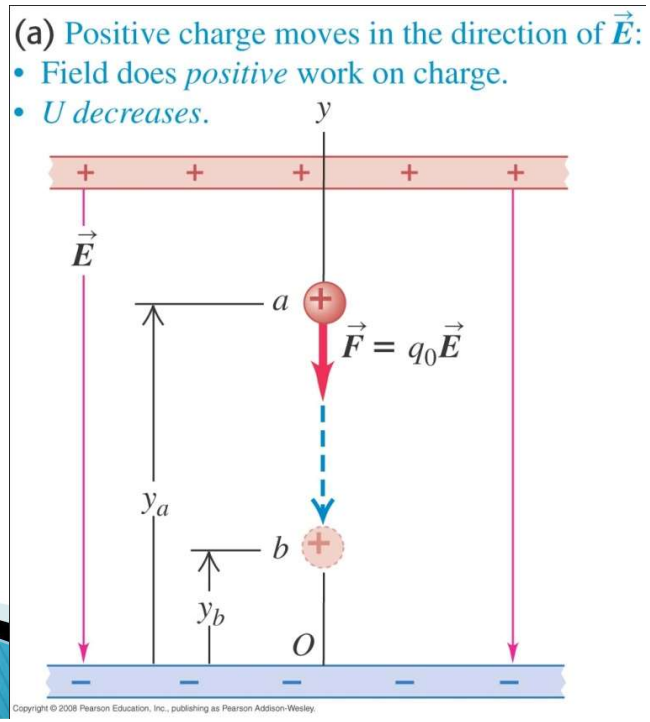


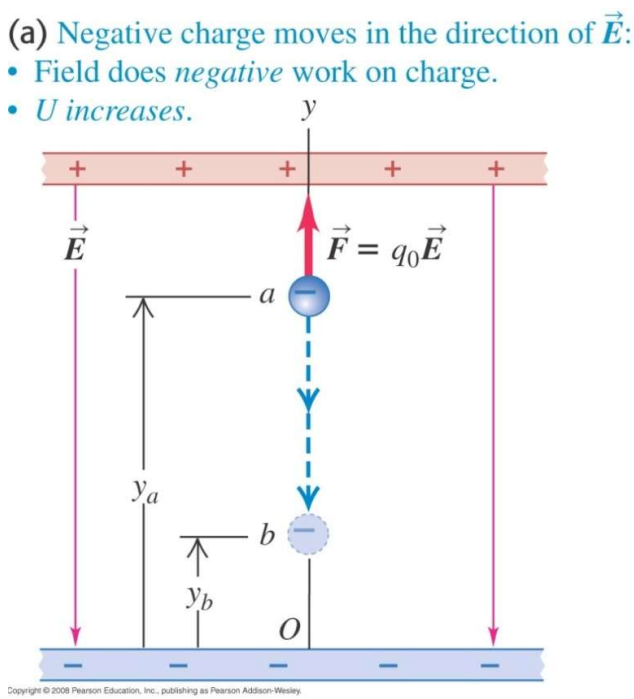
- **The electric potential energy is** $q_0 E y$.

Electric potential energy of point charges

- If the charge is positive, its electric potential energy decreases if it moves in the direction of the field.

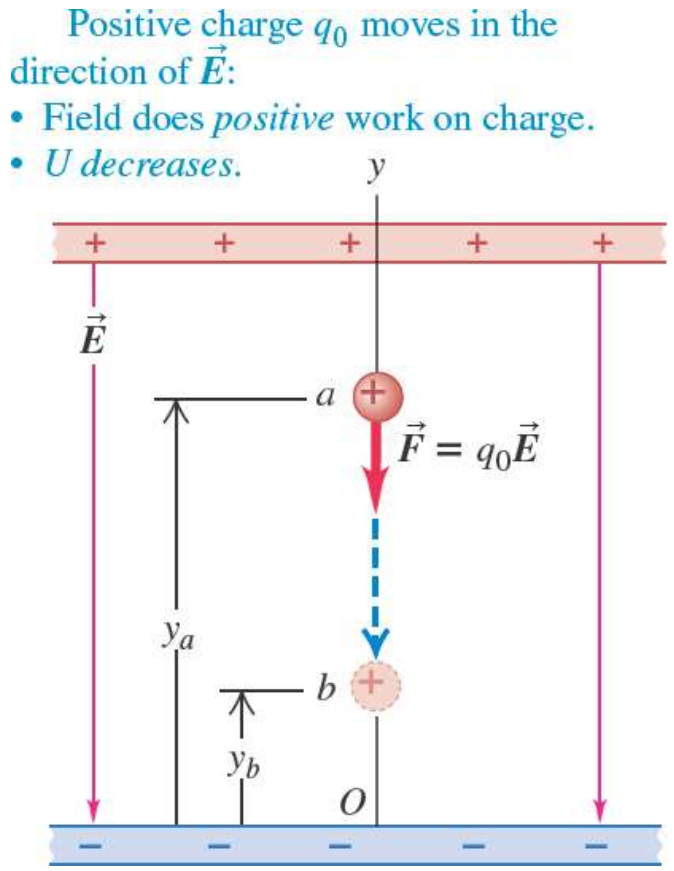
- If the charge is negative, it electric potential energy increases if it moves in the direction of the field.





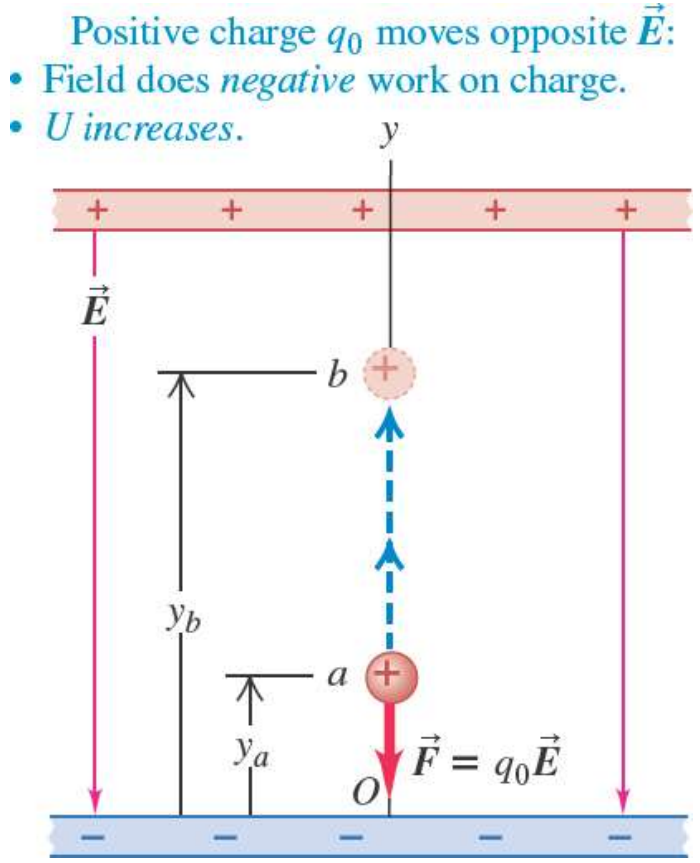
A Positive Charge Moving in a Uniform Field (1 of 2)

- If the positive charge moves in the direction of the field, the field does **positive** work on the charge.
- The potential energy **decreases.**



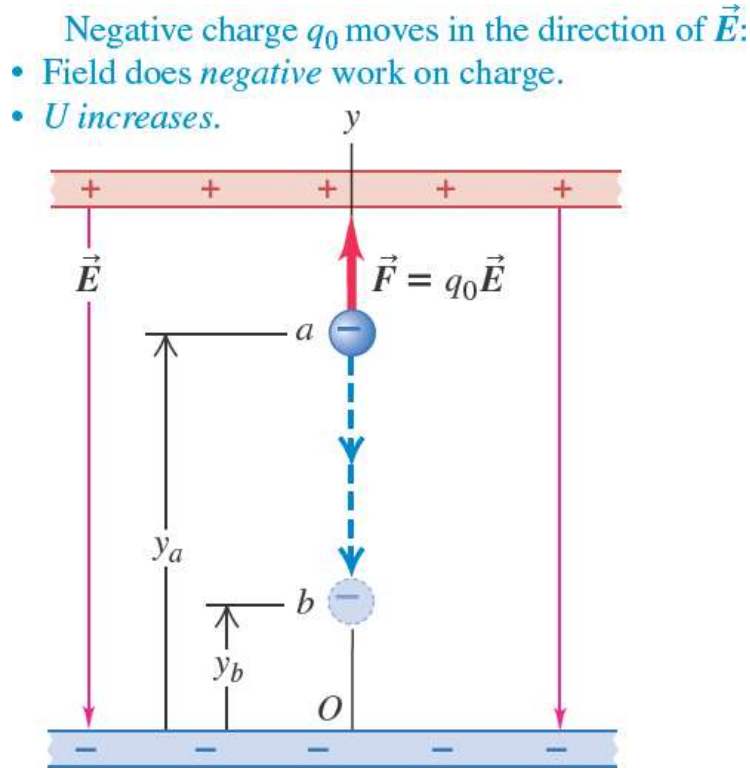
A Positive Charge Moving in a Uniform Field (2 of 2)

- If the positive charge moves opposite the direction of the field, the field does **negative** work on the charge.
- The potential energy **increases**.



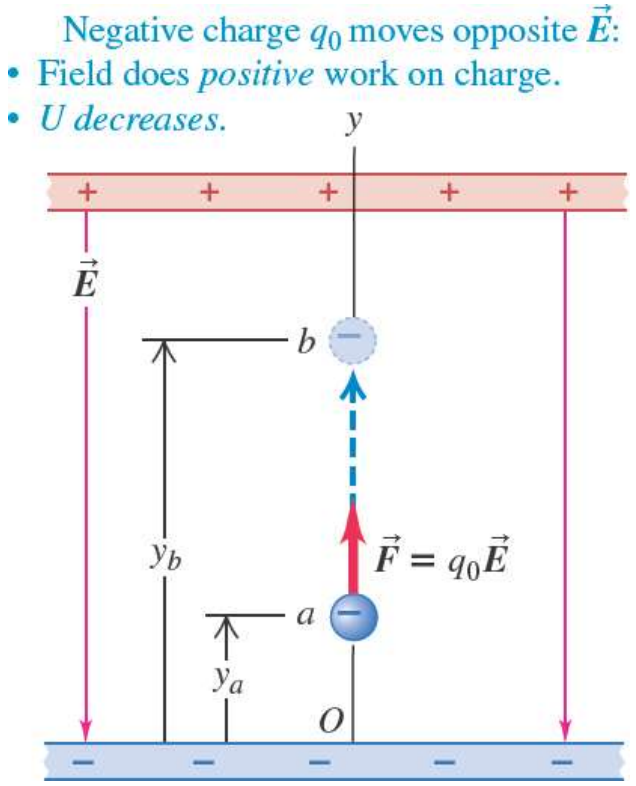
A Negative Charge Moving in a Uniform Field (1 of 2)

- If the negative charge moves in the direction of the field, the field does **negative** work on the charge.
- The potential energy **increases**.



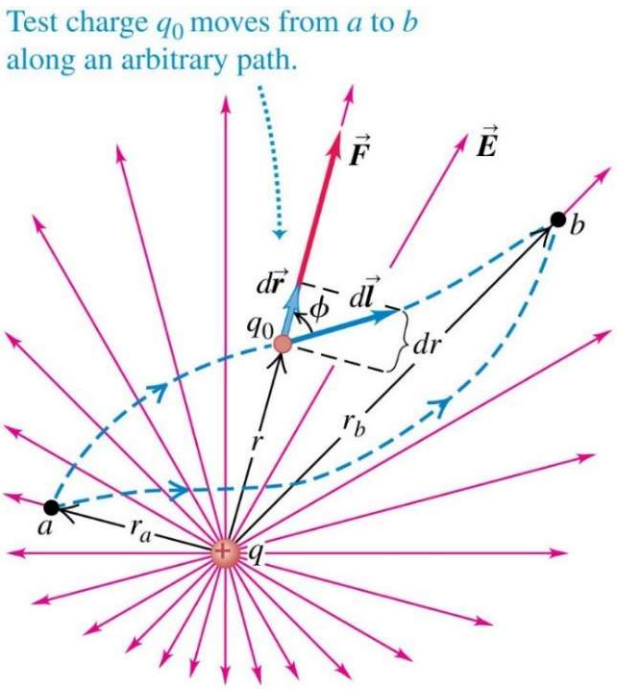
A Negative Charge Moving in a Uniform Field (2 of 2)

- If the negative charge moves opposite the direction of the field, the field does **positive** work on the charge.
- The potential energy **decreases**.



Electric Potential Energy of Two Point Charges (1 of 2)

- The work done by the electric field of one point charge on another does not depend on the path taken.
- Therefore, the electric potential energy only depends on the distance between the charges.



Potential energy – radial field

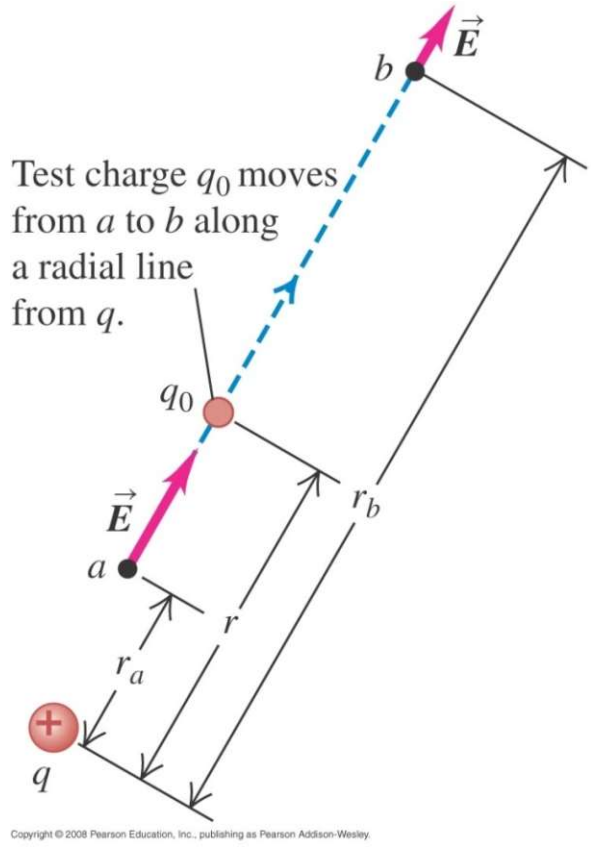
- The work to move q_0 from a to b is

$$W_{a\to b} = \int \vec{F}\cdot d\vec{l} = \int_{r_a}^{r_b} \frac{1}{4\pi\varepsilon_o}\frac{qq_0}{r^2}dr$$

$$= \frac{qq_0}{4\pi\varepsilon_o}\left(\frac{1}{r_a} - \frac{1}{r_b}\right)$$

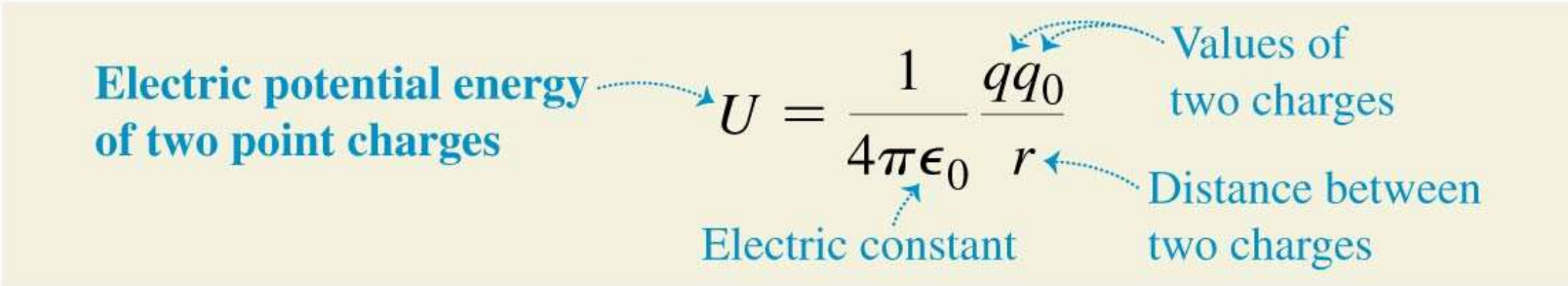
$$= U_a - U_b$$

$$\text{where } U_a = \frac{qq_0}{4\pi\varepsilon_o r_a} \quad \text{and} \quad U_b = \frac{qq_0}{4\pi\varepsilon_o r_b}$$



Electric Potential Energy of Two Point Charges (2 of 2)

- The electric potential energy of two point charges only depends on the distance between the charges.



- This equation is valid no matter what the signs of the charges are.
- Potential energy is defined to be zero when the charges are infinitely far apart.

Potential energy – radial field

- In general, at any point r from q_0, the **electric potential energy** is

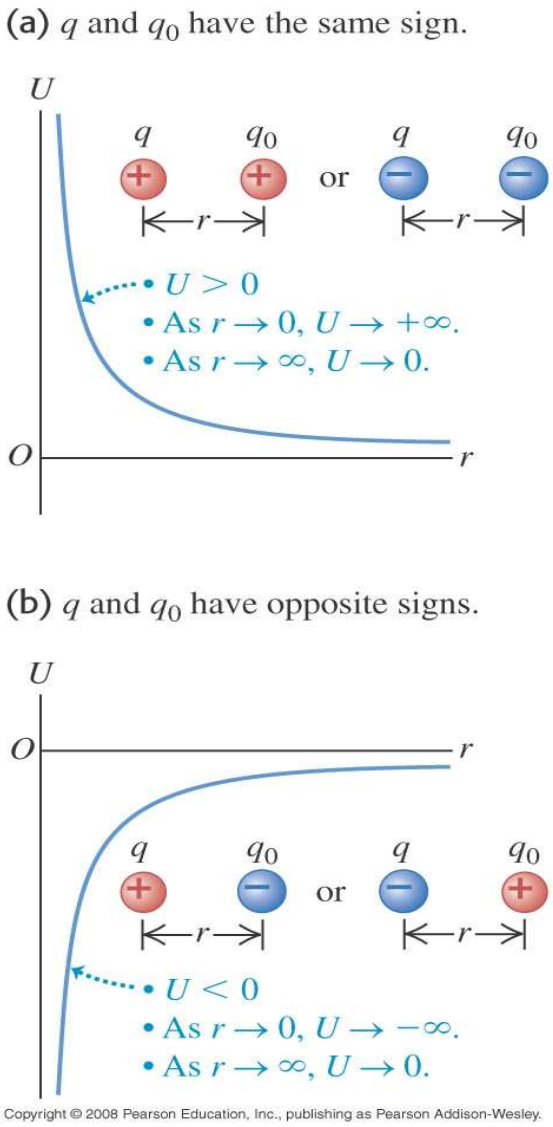
$$U = \frac{qq_0}{4\pi\varepsilon_o r}$$

- If q and q_0 have the same sign, U > 0

$$\text{As } r \to 0, U \to +\infty$$
$$\text{As } r \to \infty, U \to 0$$

- If q and q_0 have opposite signs, U < 0

$$As\ r \to 0, U \to -\infty$$
$$\text{As } r \to \infty, U \to 0$$



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Potential energy due to several point charges

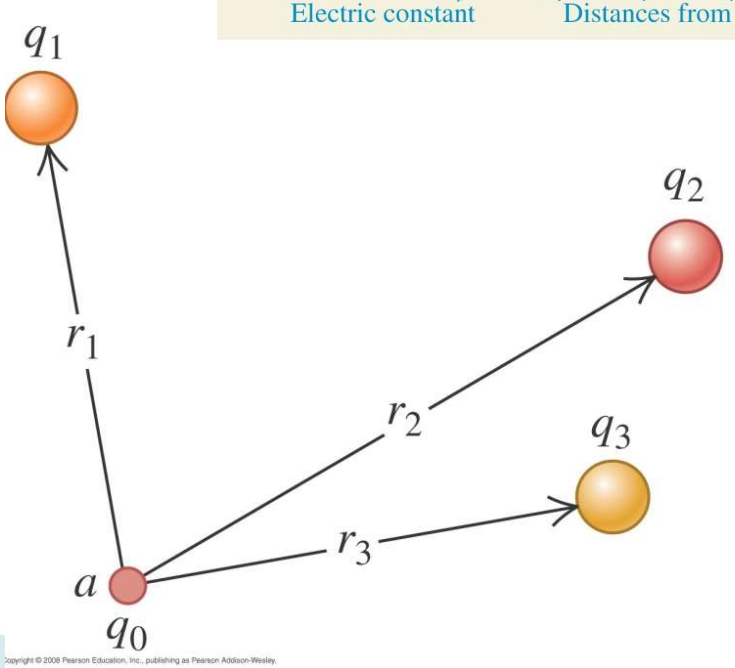
- The electric potential energy of q_0 is the algebraic sum of all the electric potential energy, i.e.

$$U = U_1 + U_2 + U_3 + \ldots$$
$$= \frac{q_0}{4\pi\varepsilon_o}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} \ldots\ldots\right)$$

Electric potential energy of point charge q_0 and collection of charges $q_1, q_2, q_3, \ldots$

$$U = \frac{q_0}{4\pi\epsilon_0}\left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \cdots\right) = \frac{q_0}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i}$$

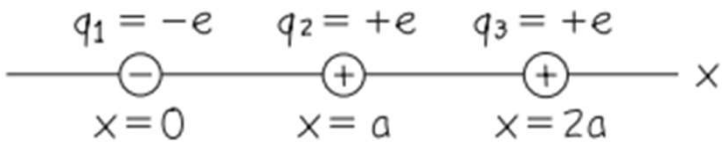
Electric constant — Distances from q_0 to $q_1, q_2, q_3, \ldots$



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Example

Two point charges are located on the x-axis, $q_1 = -e$ at $x = 0$ and $q_2 = +e$ at $x = a$. (a) Find the work that must be done by an external force to bring a third point charge $q_3 = +e$ from infinity to $x = 2a$. (b) Find the total potential energy of the system of three charges.



EXECUTE: (a) The work W equals the difference between (i) the potential energy U associated with q_3 when it is at $x = 2a$ and (ii) the potential energy when it is infinitely far away. The second of these is zero, so the work required is equal to U. The distances between the charges are $r_{13} = 2a$ and $r_{23} = a$, so

$$W = U = \frac{q_3}{4\pi\epsilon_0}\left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}}\right) = \frac{+e}{4\pi\epsilon_0}\left(\frac{-e}{2a} + \frac{+e}{a}\right) = \frac{+e^2}{8\pi\epsilon_0 a}$$

This is positive, just as we should expect. If we bring q_3 in from infinity along the $+x$-axis, it is attracted by q_1 but is repelled more strongly by q_2. Hence we must do positive work to push q_3 to the position at $x = 2a$.

(b) the total potential energy of the three-charge system is

$$U = \frac{1}{4\pi\epsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}} = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$= \frac{1}{4\pi\epsilon_0}\left[\frac{(-e)(e)}{a} + \frac{(-e)(e)}{2a} + \frac{(e)(e)}{a}\right] = \frac{-e^2}{8\pi\epsilon_0 a}$$

EVALUATE: Our negative result in part (b) means that the system has lower potential energy than it would if the three charges were infinitely far apart. An external force would have to do *negative* work to bring the three charges from infinity to assemble this entire arrangement and would have to do *positive* work to move the three charges back to infinity.

Electric potential

- Electric potential (V) is defined as the potential energy per unit charge associated with the test charge q_0 at that point.

$$\mathrm{V} = \frac{U}{q_0}$$

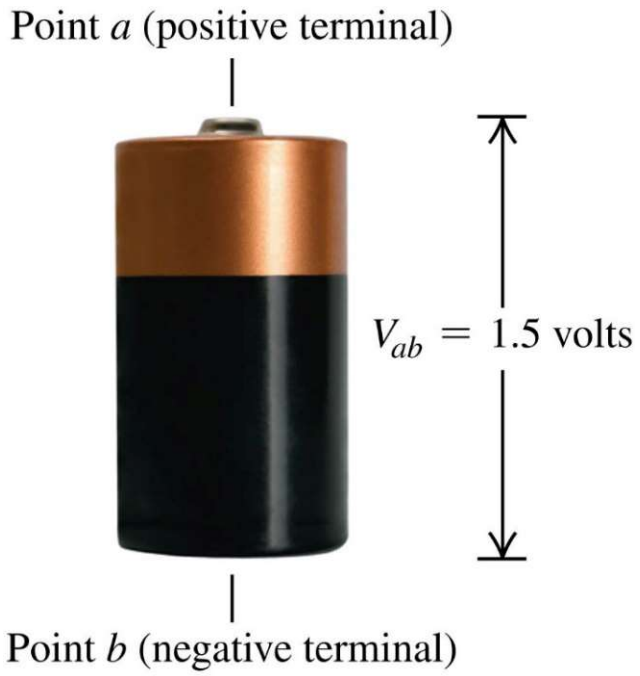
- The SI unit of potential is joule per coulomb (J/C) or volt (V).

- For a radial field, the potential at any point distance r away is

$$V = \frac{U}{q_o} = \frac{q}{4\pi\varepsilon_o r}$$

Electric Potential (1 of 2)

- **Potential is potential energy per unit charge.**
- The potential of *a* with respect to *b* ($V_{ab} = V_a - V_b$) equals the work done by the electric force when a *unit* charge moves from *a* to *b.*



Electric Potential (2 of 2)

- The potential due to a single point charge is:

Electric potential due to a point charge

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

q: Value of point charge

r: Distance from point charge to where potential is measured

ϵ_0: Electric constant

- Like electric field, potential is independent of the test charge that we use to define it.
- For a collection of point charges:

Electric potential due to a collection of point charges

$$V = \frac{1}{4\pi\epsilon_0}\sum_i \frac{q_i}{r_i}$$

q_i: Value of ith point charge

r_i: Distance from ith point charge to where potential is measured

ϵ_0: Electric constant

Electric Potential (2 of 2)

Electric potential due to a continuous distribution of charge

$$V = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r}$$

Integral over charge distribution

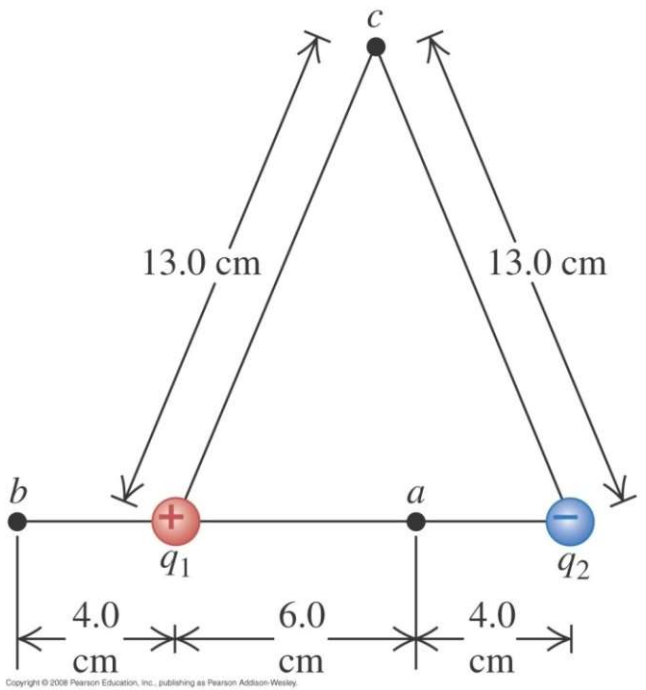
Charge element

Distance from charge element to where potential is measured

Electric constant

Example 7

An electric dipole consists of two point charges, $q_1 = +12$ nC and $q_2 = -12$ nC, placed 10 cm apart. Compute the potentials at point a, b and c by adding potentials due to either charge.



The electron-volt

- From $V = \frac{U}{q_0}$, we have $U = q_o V$

- If $q_0 = 1.6 \times 10^{-19}$ coulomb (charge of an electron) and V is 1 volt, then

  $U = q_0 V = 1.6 \times 10^{-19}$ coulomb × 1 volt = 1.6×10^{-19} J.

- By definition, this amount of energy is called 1 electron-volt (1eV).

- 1 eV = 1.6×10^{-19} J.

- The unit of eV is widely used in atomic physics.

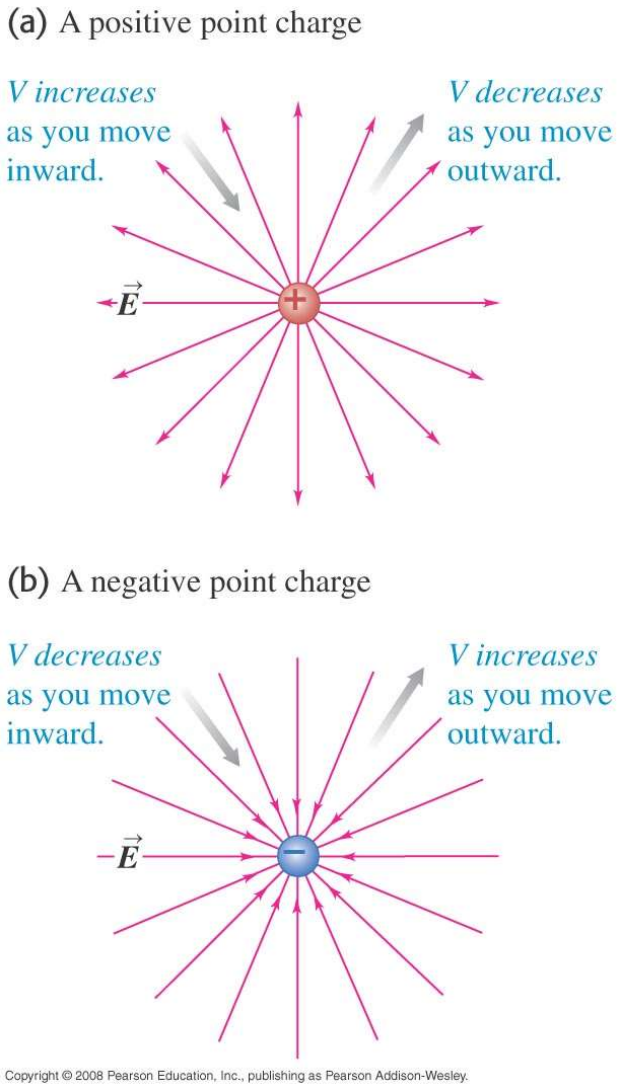
Finding potential from electric field

$$W_{a\to b} = \int_a^b \vec{F}.d\vec{l} = \int_a^b q_0 \vec{E}.d\vec{l}$$

$$V_a - V_b = \frac{W_{a\to b}}{q_0} = \int_a^b \vec{E}.d\vec{l}$$

- For a point charge $\vec{E} = \frac{q}{4\pi\varepsilon_o r^2}\hat{r}$

- Since the field is radial, $d\vec{l} = d\vec{r}$ the integral is straight forward.



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Electric potential gradient

$$V_a - V_b = \int_a^b \vec{E}.d\vec{l} = \int_a^b E\cos\phi\, dl$$

- If the field is uniform and the displacement of the charged particle is in the direction of the field, $\phi = 0$, then

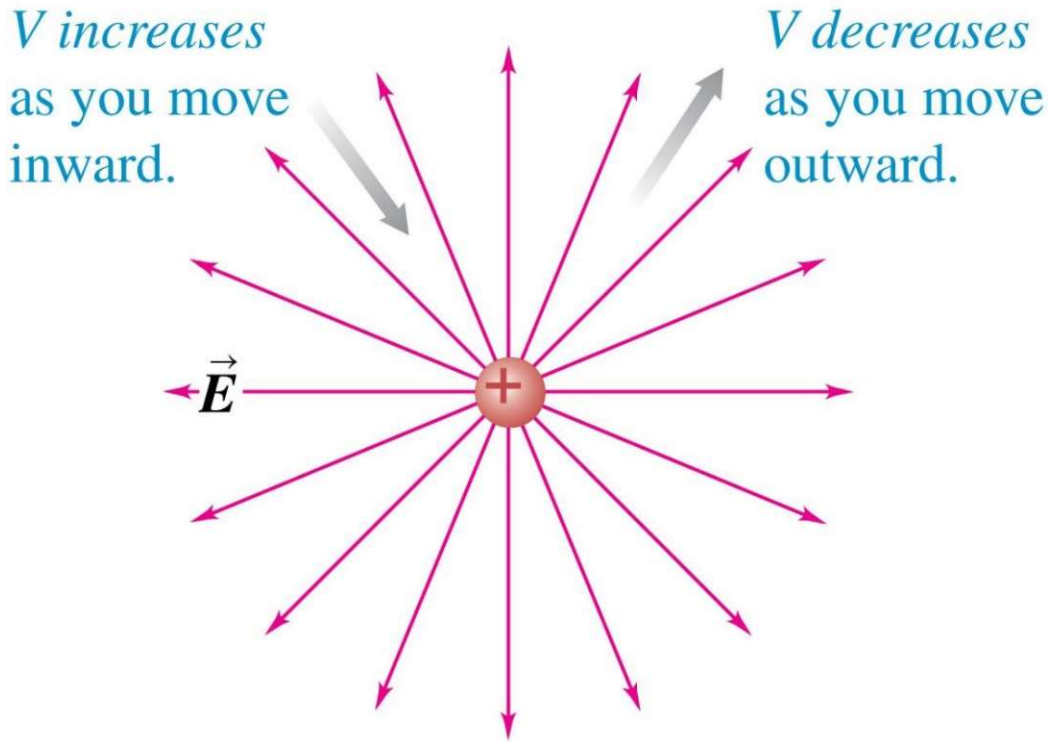
$V_a - V_b = Ed$ where d is the distance between a and b

$E = -\frac{V_b - V_a}{d} = negative$ of potential gradient

- Hence, another unit of E is V/m.

Finding Electric Potential from the Electric Field

- If you move in the direction of the electric field, the electric potential **decreases**, but if you move opposite the field, the potential **increases.**



Electric Potential and Electric Field

- Moving with the direction of the electric field means moving in the direction of decreasing V, and vice versa.
- To move a unit charge slowly against the electric force, we must apply an external force per unit charge equal and opposite to the electric force per unit charge.
- The electric force per unit charge is the electric field.
- The potential difference $V_a - V_b$ equals the work done per unit charge by this external force to move a unit charge from b to a:

$$V_a - V_b = -\int_b^a \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}}$$

$$V_f - V_i = -\int_i^f \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}}$$

- The unit of electric field can be expressed as 1 N/C = 1 V/m.

Example 8

A proton (charge e = 1.602 x 10^{-19} C) moves in a straight line from point a to point b inside a linear accelerator, a total distance d = 0.50 m. The electric field is uniform along this line, with magnitude E = 1.5 x 10^{7} V/m in the direction from a to b. Determine

a) the force on the proton

b) the work done on it by the field

c) the potential difference $V_a - V_b$

Example 9

An electric potential difference of 2000 V is applied to a pair of plates that is separated by 1 cm. If a particle of charge 6.40×10^{-19} C is released from the upper plate

a) What is the electric field (that is uniform) in the region between the plates?

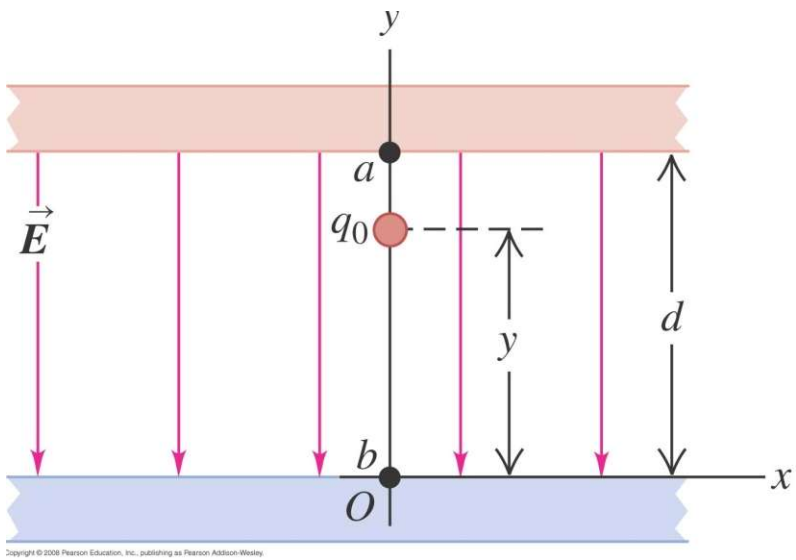
b) If the mass of the particle is 1.92×10^{-16} kg, what is the speed of the particle as it reaches the bottom plate?

$$a)\ E = \frac{V}{d} = \frac{2000}{0.01} = 200000\ \mathrm{V/m}$$

$$b)\ W = qV = \frac{1}{2}mv^2$$

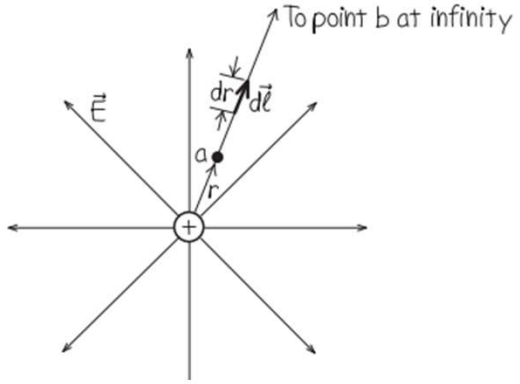
$$v^2 = \frac{2qV}{m} = \frac{2 \times 6.40 \times 10^{-19} \times 2000}{1.92 \times 10^{-1}}$$

$$v = 3.7\ \mathrm{m/s}$$



Example

Find the potential at a distance *r* from a point charge *q*, given the Electric field.



$$V - 0 = V = \int_r^\infty \vec{E} \cdot d\vec{l}$$

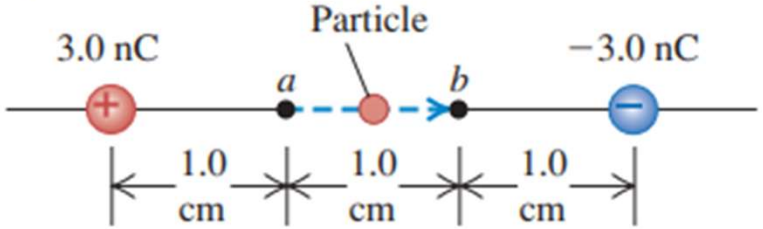
$$= \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} \cdot \hat{\mathbf{r}} dr = \int_r^\infty \frac{q}{4\pi\epsilon_0 r^2} dr$$

$$= -\frac{q}{4\pi\epsilon_0 r}\bigg|_r^\infty = 0 - \left(-\frac{q}{4\pi\epsilon_0 r}\right)$$

$$V = \frac{q}{4\pi\epsilon_0 r}$$

Example

In the fig, a dust particle with mass $m = 5.0 \times 10^{-9}$ kg $= 5.0\ \mu$g and charge $q_0 = 2.0$ nC starts from rest and moves in a straight line from point a to point b. What is its speed v at point b?



We have $K_a = 0$ and $K_b = \frac{1}{2}mv^2$. We substitute these and our expressions for U_a and U_b into the energy-conservation equation, then solve for v. We find

$$0 + q_0V_a = \tfrac{1}{2}mv^2 + q_0V_b$$

$$v = \sqrt{\frac{2q_0(V_a - V_b)}{m}}$$

We calculate the potentials using $V = q/4\pi\epsilon_0 r$:

$$V_a = (9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{3.0 \times 10^{-9}\ \text{C}}{0.010\ \text{m}} + \frac{(-3.0 \times 10^{-9}\ \text{C})}{0.020\ \text{m}}\right) = 1350\ \text{V}$$

$$V_b = (9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)\left(\frac{3.0 \times 10^{-9}\ \text{C}}{0.020\ \text{m}} + \frac{(-3.0 \times 10^{-9}\ \text{C})}{0.010\ \text{m}}\right) = -1350\ \text{V}$$

$$V_a - V_b = (1350\ \text{V}) - (-1350\ \text{V}) = 2700\ \text{V}$$

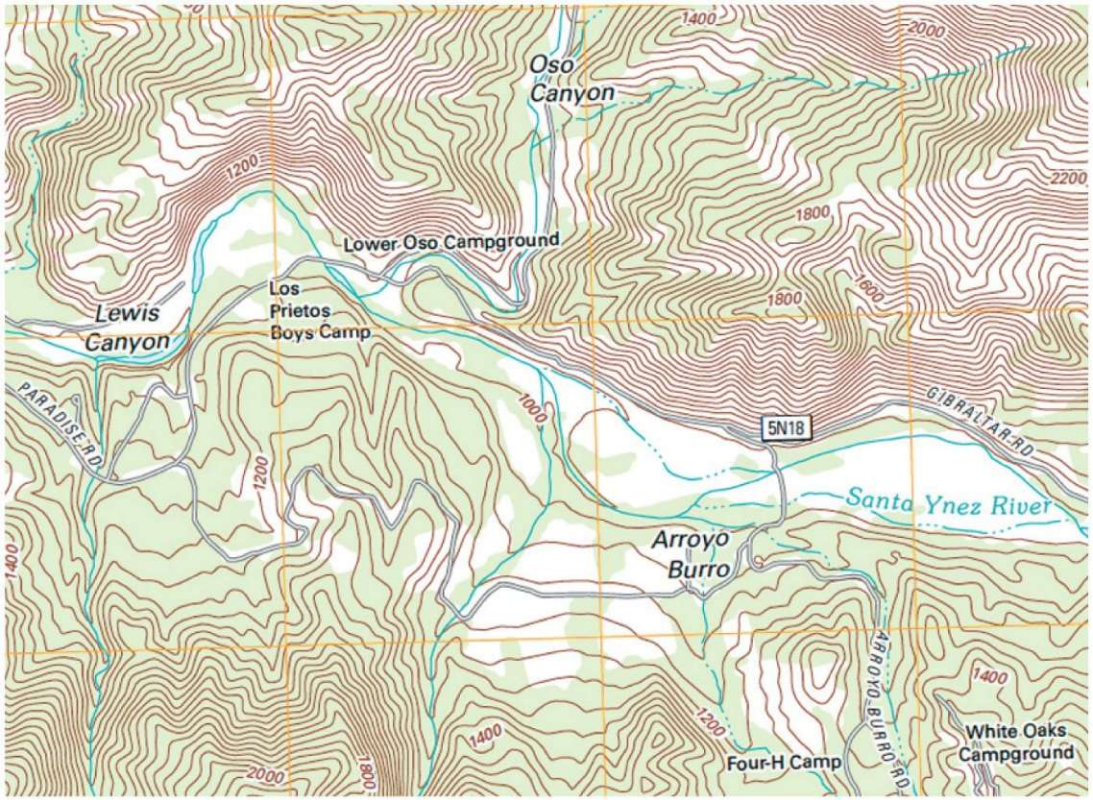
Finally,

$$v = \sqrt{\frac{2(2.0 \times 10^{-9}\ \text{C})(2700\ \text{V})}{5.0 \times 10^{-9}\ \text{kg}}} = 46\ \text{m/s}$$

EVALUATE: Our result makes sense: The positive test charge speeds up as it moves away from the positive charge and toward the negative charge. To check unit consistency in the final line of the calculation, note that 1 V = 1 J/C, so the numerator under the radical has units of J or $\text{kg}\cdot\text{m}^2/\text{s}^2$.

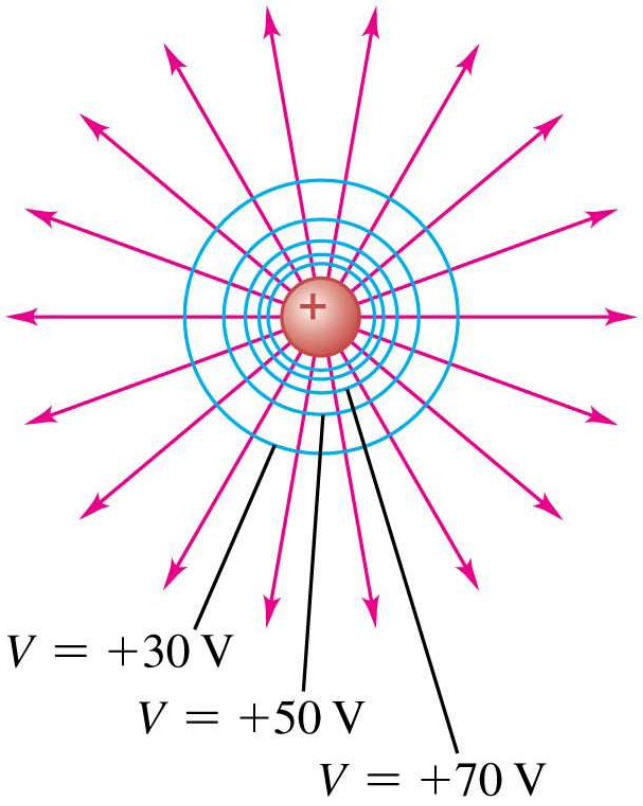
Equipotential Surfaces

- Contour lines on a topographic map are curves of constant elevation and hence of constant gravitational potential energy.



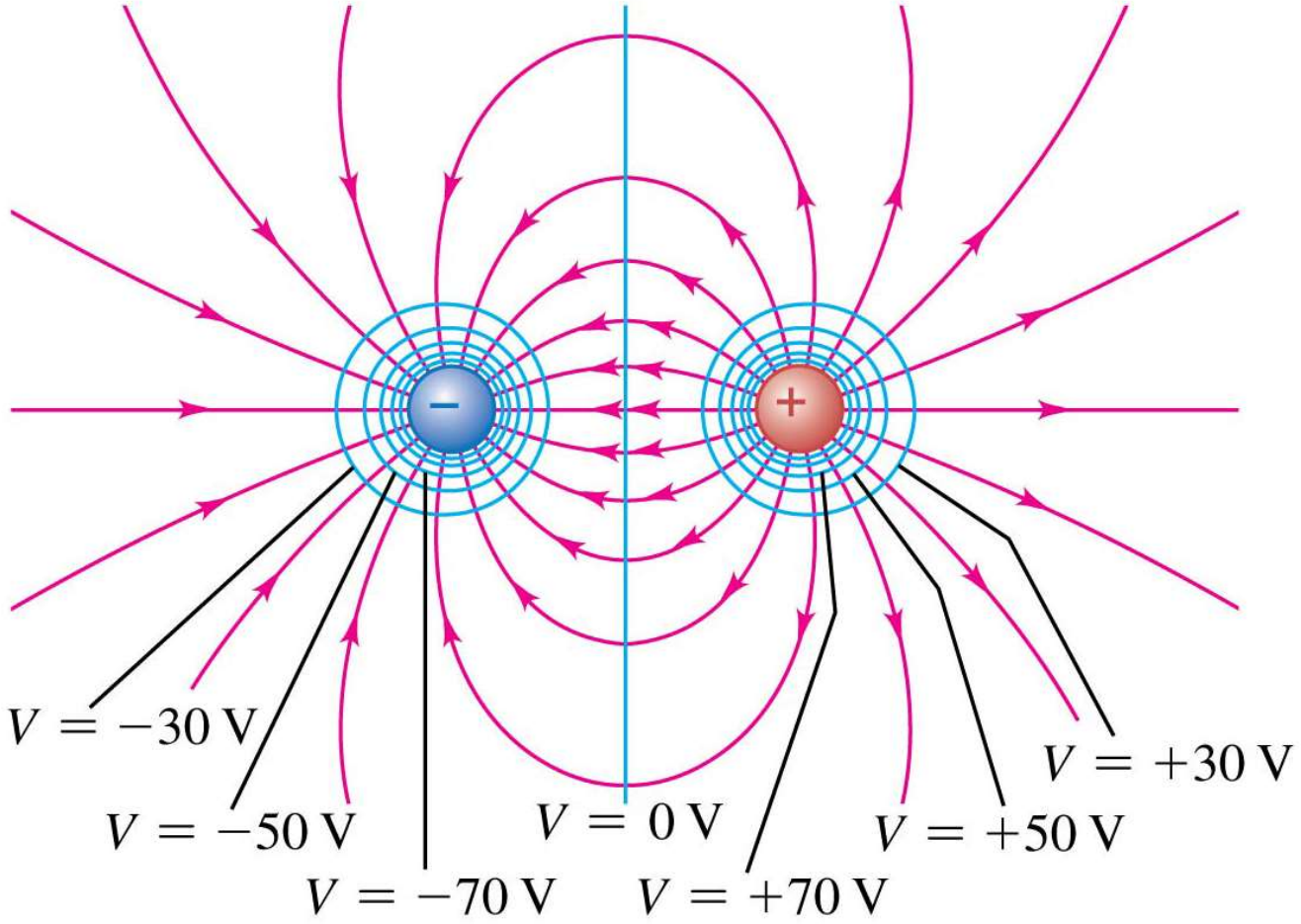
Equipotential Surfaces and Field Lines

- An **equipotential surface** is a surface on which the electric potential is the same at every point.
- Field lines and equipotential surfaces are always mutually perpendicular.
- Shown are cross sections of equipotential surfaces (blue lines) and electric field lines (red lines) for a single positive charge.

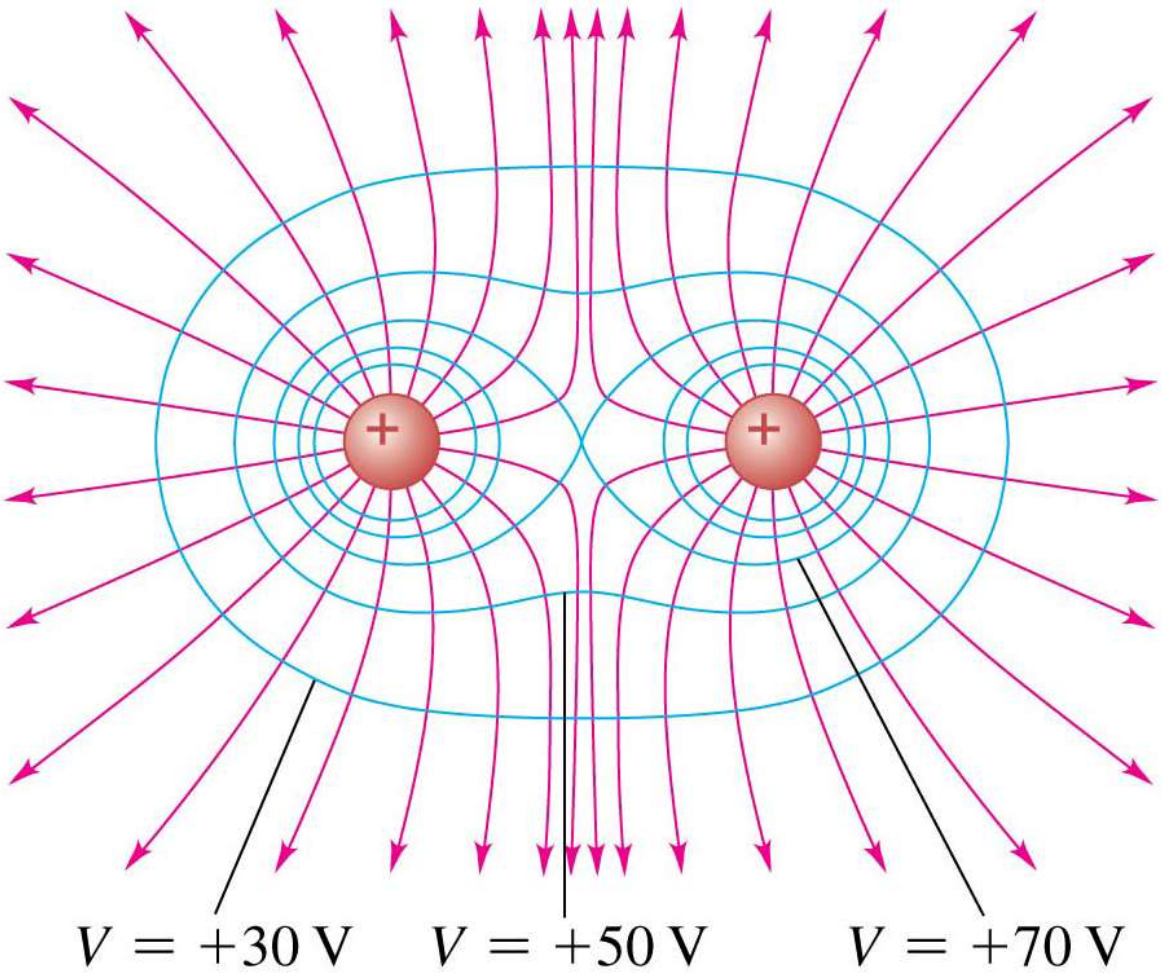


- Video Tutor Demonstration: Charged Conductor with Teardrop Shape

Equipotential Surfaces and Field Lines for a Dipole

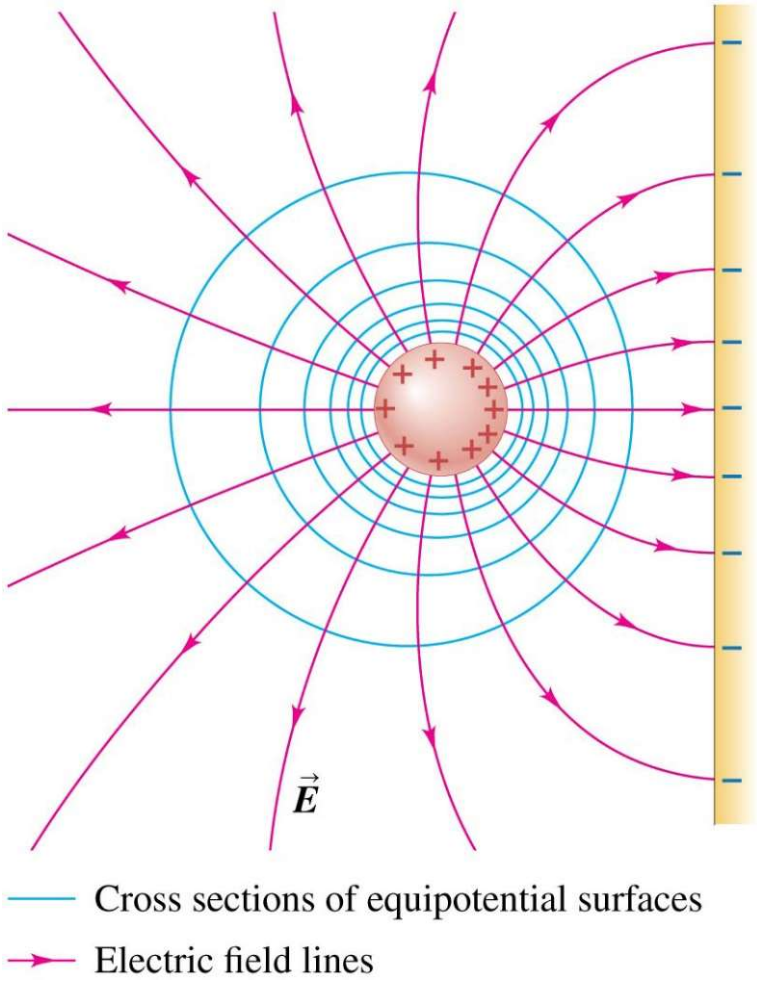


Field and Potential of Two Equal Positive Charges



Equipotentials and Conductors (1 of 2)

- When all charges are at rest:
  - the surface of a conductor is always an equipotential surface.
  - the electric field just outside a conductor is always perpendicular to the surface.

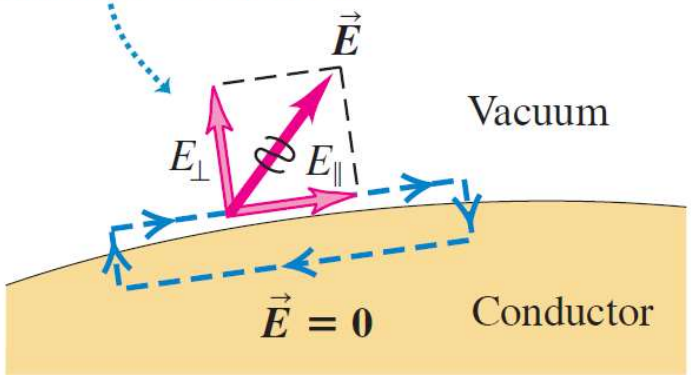


Equipotentials and Conductors (2 of 2)

- If the electric field had a tangential component at the surface of a conductor, a net amount of work would be done on a test charge by moving it around a loop as shown here—which is impossible because the electric force is conservative.

An impossible electric field

If the electric field just outside a conductor had a tangential component $E_{\parallel}$, a charge could move in a loop with net work done. Hence the field just outside a conductor can have only a perpendicular component.



Electric Field and Potential Gradient

$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l}$$

$$V_a - V_b = \int_b^a dV = -\int_a^b dV$$

where dV is the infinitesimal change of potential accompanying an infinitesimal element $d\vec{l}$ of the path from b to a.

$$-\int_a^b dV = \int_a^b \vec{E} \cdot d\vec{l}$$

These two integrals must be equal for *any* pair of limits a and b, and for this to be true the *integrands* must be equal. Thus, for *any* infinitesimal displacement $d\vec{l}$,

$$-dV = \vec{E} \cdot d\vec{l}$$

To interpret this expression, we write $\vec{E}$ and $d\vec{l}$ in terms of their components: $\vec{E} = \hat{i}\,E_x + \hat{j}\,E_y + \hat{k}\,E_z$ and $d\vec{l} = \hat{i}\,dx + \hat{j}\,dy + \hat{k}\,dz$. Then we have

$$-dV = E_x dx + E_y dy + E_z dz$$

Suppose the displacement is parallel to the x-axis, so $dy = dz = 0$. Then $-dV = E_x dx$ or $E_x = -(dV/dx)_{y,\,z\ \text{constant}}$, where the subscript reminds us that only x varies in the derivative; recall that V is in general a function of x, y, and z. But this is just what is meant by the partial derivative $\partial V/\partial x$. The y- and z-components of $\vec{E}$ are related to the corresponding derivatives of V in the same way, so we have

$$E_x = -\frac{\partial V}{\partial x} \qquad E_y = -\frac{\partial V}{\partial y} \qquad E_z = -\frac{\partial V}{\partial z}$$

(components of $\vec{E}$ in terms of V)

Electric Field and Potential Gradient

$$\vec{E} = -\left(\hat{i}\frac{\partial V}{\partial x} + \hat{j}\frac{\partial V}{\partial y} + \hat{k}\frac{\partial V}{\partial z}\right) \qquad (\vec{E} \text{ in terms of } V)$$

In vector notation the following operation is called the **gradient** of the function f:

$$\vec{\nabla} f = \left(\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) f$$

The operator denoted by the symbol $\vec{\nabla}$ is called "grad" or "del." Thus in vector notation,

$$\vec{E} = -\vec{\nabla} V$$

This is read "$\vec{E}$ is the negative of the gradient of V" or "$\vec{E}$ equals negative grad V." The quantity $\vec{\nabla} V$ is called the *potential gradient.*

Example

If it is given that for a ring of charge with radius a and total charge Q, the potential at a point P on the ring's symmetry axis a distance x from the center is

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{x^2 + a^2}}$$

Find the electric field at P.

The x-component of the electric field is

$$E_x = -\frac{\partial V}{\partial x} = \frac{1}{4\pi\epsilon_0} \frac{Qx}{(x^2 + a^2)^{3/2}}$$

Electric Potential for continuous charge distribution

Electric potential due to a continuous distribution of charge

$$V = \frac{1}{4\pi\epsilon_0}\int \frac{dq}{r}$$

Integral over charge distribution

Charge element

Distance from charge element to where potential is measured

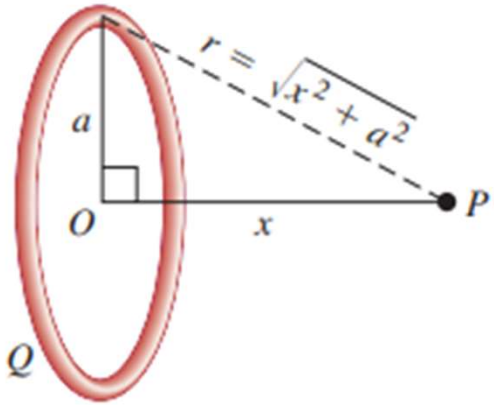
Electric constant

Example

Find the potential due to a uniform charge distribution, Q of a ring of radius a at a point on the axis of the ring.

EXECUTE: The fig shows that the distance from each charge element dq to P is $r = \sqrt{x^2 + a^2}$. Hence we can take the factor $1/r$ outside the integral in Eq. (23.16), and

$$V = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r} = \frac{1}{4\pi\epsilon_0}\frac{1}{\sqrt{x^2 + a^2}}\int dq = \frac{1}{4\pi\epsilon_0}\frac{Q}{\sqrt{x^2 + a^2}}$$



EVALUATE: When x is much larger than a, our expression for V becomes approximately $V = Q/4\pi\epsilon_0 x$, which is the potential at a distance x from a point charge Q.

Example

Positive electric charge Q is distributed uniformly along a line of length $2a$ lying along the y-axis between $y = -a$ and $y = +a$. Find the electric potential at a point P on the x-axis at a distance x from the origin.

EXECUTE: As before, the element of charge dQ corresponding to an element of length dy on the rod is $dQ = (Q/2a)dy$. The distance from dQ to P is $\sqrt{x^2 + y^2}$, so the contribution dV that the charge element makes to the potential at P is

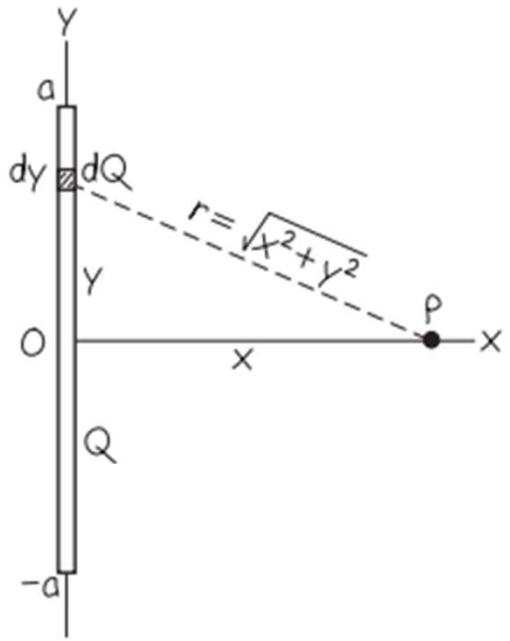
$$dV = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\frac{dy}{\sqrt{x^2 + y^2}}$$

To find the potential at P due to the entire rod, we integrate dV over the length of the rod from $y = -a$ to $y = a$:

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\int_{-a}^{a}\frac{dy}{\sqrt{x^2 + y^2}}$$

You can look up the integral in a table. The final result is

$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{2a}\ln\left(\frac{\sqrt{a^2 + x^2} + a}{\sqrt{a^2 + x^2} - a}\right)$$



Example

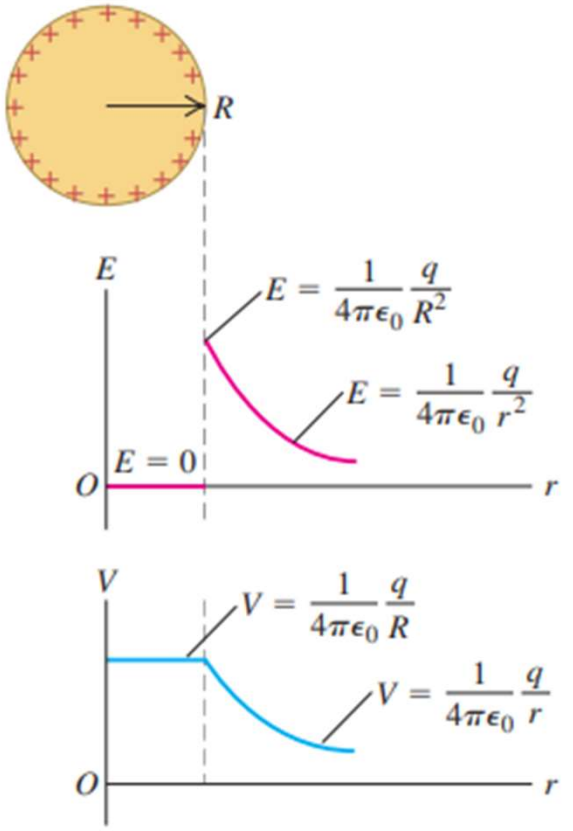
A solid conducting sphere of radius R has a total charge q. Find the electric potential everywhere, both outside and inside the sphere.

EXECUTE: [illegible] the field *outside* the sphere is th same as if the sphere were removed and replaced by a point charg q. We take $V = 0$ at infinity, as we did for a point charge. Then th potential at a point outside the sphere at a distance r from its cente is the same as that due to a point charge q at the center:

$$V = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

The potential at the surface of the sphere is $V_{\text{surface}} = q/4\pi\epsilon_0 R$.

Inside the sphere, $\vec{E}$ is zero everywhere. Hence no work done on a test charge that moves from any point to any other poi inside the sphere. This means that the potential is the same every point inside the sphere and is equal to its value $q/4\pi\epsilon_0 R$ the surface.



Example

Find the potential at a distance r from a very long line of charge with linear charge density (charge per unit length) λ.

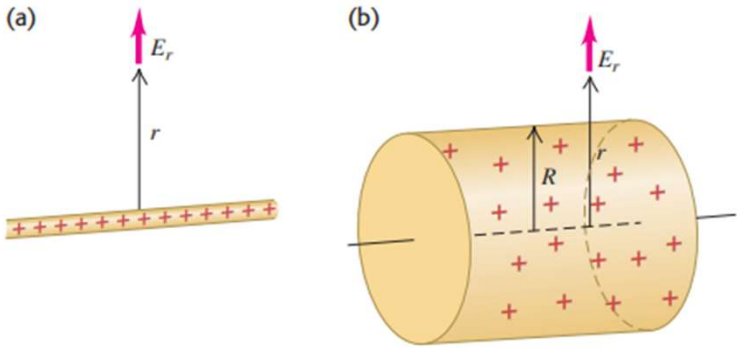
$$V_a - V_b = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b E_r\, dr = \frac{\lambda}{2\pi\epsilon_0} \int_{r_a}^{r_b} \frac{dr}{r} = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_b}{r_a}$$

If we take point b at infinity and set $V_b = 0$, we find that V_a is *infinite* for any finite distance r_a from the line charge: $V_a = (\lambda/2\pi\epsilon_0) \ln(\infty/r_a) = \infty$. This is *not* a useful way to define V for this problem! The difficulty is that the charge distribution itself extends to infinity.

we set $V_b = 0$ at point b at an arbitrary but *finite* radial distance r_0. Then the potential $V = V_a$ at point a at a radial distance r is given by $V - 0 = (\lambda/2\pi\epsilon_0) \ln(r_0/r)$, or

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_0}{r}$$

Electric field outside (a) a long, positively charged wire and (b) a long, positively charged cylinder.



of r (the distance from the cylinder axis) equal to or greater than the radius R of the cylinder. If we choose r_0 to be the cylinder radius R, so that $V = 0$ when $r = R$, then at any point for which $r > R$,

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R}{r}$$

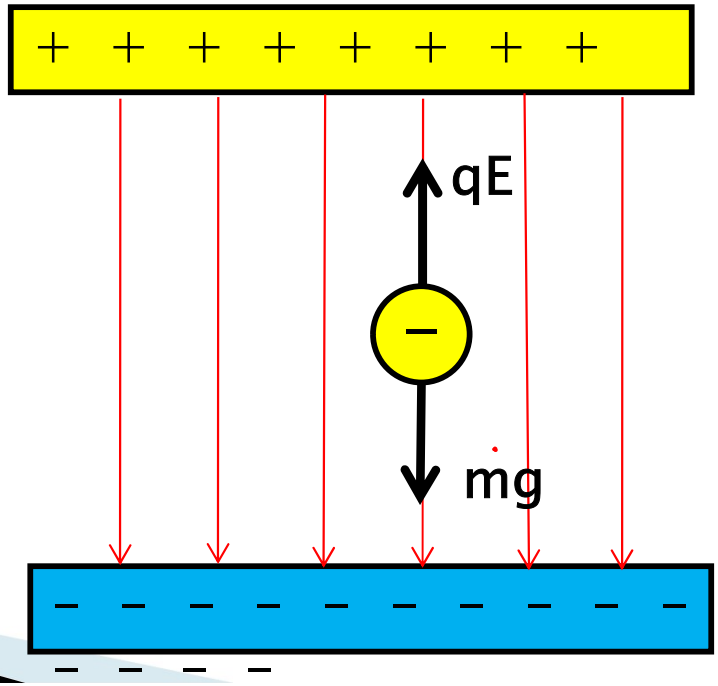
Inside the cylinder, $\vec{E} = \mathbf{0}$, and V has the same value (zero) as on the cylinder's surface.

Millikan oil drop experiment

- A negatively charged oil drop is held stationary between two charged plates.

$$qE = mg, \text{ or } q = mg/E$$

- If we know the mass of the oil drop, and the electric field, we can calculate the charge on the oil drop!



Example 10

In a particular Millikan oil drop apparatus, the plates are 2 cm apart. An electric field strength field of 5×10^4 N/C caused an oil drop to become stationary. The oil used has a density of 885 kgm^{-3}.

a) What is the potential difference that is applied to the plates?

b) The oil drop was found to have a volume of 7.24×10^{-18} m^3. What is the charge on the oil drop?

c) If the oil drop is negatively charged, how many extra electrons does it have?

$a) V = Ed = 5 \times 10^4 \times 0.02 = 1000 \text{ V}$

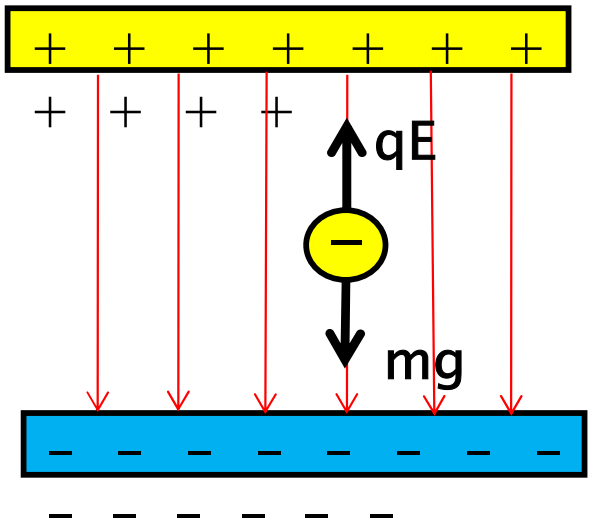
$b) m = 885 \times 7.24 \times 10^{-18}$

$c) mg = qE$

$$q = \frac{885 \times 7.24 \times 10^{-1} \times 9.81}{5 \times 10^4}$$

$$= 1.257 \times 10^{-18} \text{ C}$$

$No \text{ of electrons} = 7.85 = 8$



End of chapter